Task	Method	Originality	Elaboration
AUT	SA	3.11±1.21	3.43±1.10
	LLM Discussion	3.55±1.07	2.78±1.09
	BILLY (Ours)	3.67±1.11	4.08±0.89
INSTANCES	SA	2.53±1.15	2.55±1.16
	LLM Discussion	3.45±1.19	3.04±1.08
	BILLY (Ours)	3.75±1.38	3.96±0.90
SCIENTIFIC	SA	3.01±1.37	3.86±1.07
	LLM Discussion	3.30±1.16	2.70±0.93
	BILLY (Ours)	3.47±1.18	4.23±0.85
SIMILARITIES	SA	3.15±1.40	3.79±0.99
	LLM Discussion	2.79±1.11	2.33±1.13
	BILLY (Ours)	3.75±1.09	4.17±1.07

Table 3: **Human Evaluation Results.** The highest scores are highlighted in purple, while the second-highest are in blue. Across all benchmarks, BILLY (Ours) consistently achieves the strongest Originality and Elaboration scores.

Correlation	Originality	Elaboration
Spearman	0.73	0.43
Pearson	0.66	0.40

Table 4: **Correlation Between Human and LLM Judges.** Spearman and Pearson correlations between LLM-based evaluations and average human evaluations for Originality and Elaboration.

particularly strong gains in Elaboration, indicating its ability to produce more detailed responses. Meanwhile, although LLM Discussion often improves originality over SA, its elaboration scores are consistently lower. These results suggest that direct activation steering provides a more effective and stable mechanism for eliciting creative responses.

We also calculated the Spearman's Rank Correlation Coefficient and Pearson Correlation Coefficient between our LLM-Judge and human raters, with the results show in Table 4. The results indicate a strong positive correlation for the Originality. While the correlation for the Elaboration score is more moderate, this aligns with observations reported in the LLM-Discussion (Lu et al., 2024).

5 Discussion & Analysis

Our analysis goes beyond simple confirmation of efficacy to provide deeper insights into the mechanisms of BILLY. To dissect these mechanisms, our discussion is structured around four concepts: (1) The actual token cost and inference time among different methods.(2) The relationship between the quantity of fused vectors and the resulting creative quality. (3) Qualitative nature of the generated text. (4) Layer-wise projection among different methods. The experiments in this section adopt the Llama-3.1-8B-Instruct model, which we find sufficiently representative for our analysis.

5.1 Token Cost and Inference Time

Table 5 details the average inference latency, input and output token counts per query, and estimated token cost per 10000 queries for each method. The data clearly demonstrates BILLY's superior efficiency, reducing both token usage and associated estimated cost by **more than 95%** compared to the LLM Discussion baseline. Notably, the amortized approach emerges as the most cost-effective solution, as its operational costs remain fixed regardless of query volume. We discuss the amortized cost, which is the most evident advantage of BILLY, in Section E, and Section J also provides the complete chat log of a specific agent in LLM Discussion with a single question.

Practical efficiency of BILLY is evident not only in its token cost but also in its substantially shorter inference time. We measured the wall-clock time required for each method to answer 100 AUT questions and calculated the average inference latency per query. As detailed in Table 5, BILLY is approximately **25 times faster** than LLM Discussion. This dramatic reduction in latency suggests that our approach is a practical solution for real-time creative generation.

Method	Latency (s)	Token (In / Out)[1]	Cost ($)[2]
SA	23.9	22.3 / 407.1	0.25
SA-MRP	36.8	221.2 / 861.1	0.56
LLM Discussion	513	88853 / 12922.1	25.50
BILLY (Ours)	19	62.2 / 475.6	0.30

[1] Token are per-query averages (n=10,000).
[2] Cost is the estimated price for 10000 queries calculated based on Nebius AI Studio, where the price is $0.02 / $0.06 USD per million input / output tokens.

Table 5: **Inference Time and Token Cost.** BILLY demonstrates a significant reduction in both latency and token cost compared to LLM Discussion.

5.2 Analysis of Various Vector Compositions

To investigate the relationship between the quantity of fused personas and creativity, we vary the number of merged vectors from one to seven. The results are presented in Figure 2 with the specific persona vector combinations detailed in Section F. First, any vector merge with the **creative professional** persona results in exceptionally high creative performance. Second, while increasing the

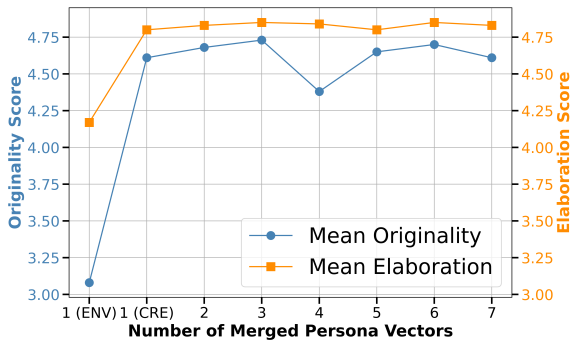

Figure 2: **Persona Vector Combinations Analysis.** Based on the default 4 vectors, we modify the combination of persona vectors from one to seven.

number of persona vectors from one, *i.e.*, 1 (CRE), to three yields a noticeable improvement, further additions from four to seven vectors do not produce additional significant gains. This suggests that the primary benefit of BILLY is not simply additive.

5.3 A Case Study on Persona Fusion

We present a case study on the *Reimagine the Hospital* task, one of the neutral questions designed for our activation extraction (Section 5.4). The results, detailed in Figure 3 and Section G, demonstrate that the individual vectors exhibit distinct functions: the environmentalist generates a pragmatic blueprint (*e.g.*, Community Gardens), while the creative professional produces an evocative, artistic term (*e.g.*, Bioluminous Biome). The merged vector demonstrates a true conceptual fusion. It retains the substantive concepts from the environmentalist but reframes them with the visionary style of the creative vector. For instance, a functional "Community Garden" is transformed into an experiential "Fractal Forest." In contrast, while prompted to be an environmentalist and creative professional, MRP uses relatively analytical words to convey the concept of sustainability rather than creative ones.

5.4 Steering Alignment via Projection

To verify how effectively our method aligns model representations with target personas, we analyze the projection of activation changes $\Delta\vec{a}^{(l)}$ in Equation (4) onto layer-specific persona vectors. This analysis is conducted using 10 neutral questions shown in Section H and 10 randomly selected questions from our evaluation dataset. The core idea is to measure the **change** in the model's activations caused by our steering vector and then calculate how much of the change occurred along the intended persona's direction.

Question: Healthcare Innovation - Redesign the future hospital experience for patients, families, and staff.

BILLY (ENV)
… **Community Gardens**: Outdoor spaces promote physical activity, connect nature, and reduce stress.

BILLY (CRE)
… **Bioluminous Biome**: Radiant forests, crystalline geodes, & iridescent orbs bathing spaces in kaleidoscopic hues.

BILLY (CRE + ENV)
… **Fractal Forests:** Ephemeral gardens grow within glass domes, bathing visitors in iridescent mosaics, awakening inner blooms.

MRP (CRE + ENV)
… **Feasibility and Sustainability**: Balance operational costs with environmental sustainability, energy efficiency, and long-term viability.

Figure 3: **Qualitative Results.** Responses generated by models that are steered by BILLY (ENV), BILLY (CRE), BILLY (CRE + ENV), and MRP (CRE + ENV).

This analysis involves two main steps. First, we define the **activation difference vector**, $\Delta\vec{a}^{(l)}$, for each layer l. This vector is the difference between the **steered activations** $\vec{a}^{(l)}_{\text{steered}}$ from Equation (3) and the **original activations** $\vec{a}^{(l)}_{\text{original}}$ from a standard forward pass without steering:

$$\Delta\vec{a}^{(l)} = \vec{a}^{(l)}_{\text{steered}} - \vec{a}^{(l)}_{\text{original}} \tag{4}$$

Second, to quantify the alignment of this change with a specific persona, we project the activation difference vector $\Delta\vec{a}^{(l)}$ onto that persona's predefined, layer-specific unit persona vector. The resulting projection value, $Projection^{(l)}_{\text{persona}}$, is calculated via the dot product:

$$Projection^{(l)}_{\text{persona}} = \Delta\vec{a}^{(l)} \cdot \frac{\vec{v}^{(l)}_{\text{persona}}}{\|\vec{v}^{(l)}_{\text{persona}}\|} \tag{5}$$

where $\vec{v}^{(l)}_{\text{persona}}$ is the persona vector for a given persona at layer l. A higher positive projection value indicates that our steering intervention more strongly shifted the model's activations along that persona's semantic axis.

Superiority of Vector Steering over Prompting. Figure 4 shows that while prompting model to be both a creative professional and an environmentalist, the projection is only positive on v_{CRE} (Figure 4a) but negative on v_{ENV} (Figure 4b). This

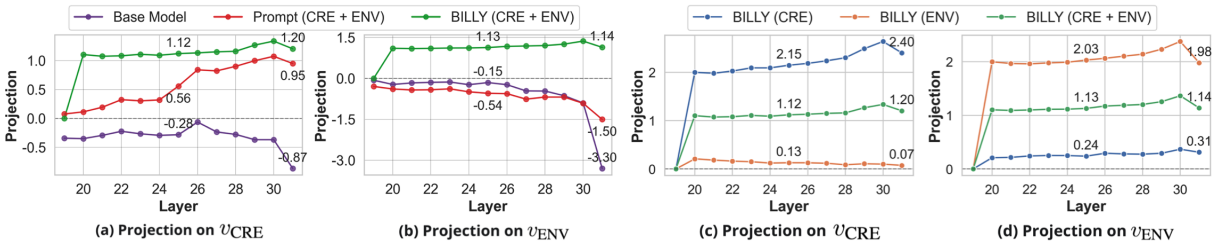

Figure 4: **Projection of Different Methods.** Projection of activation changes on the layer-specific creative professional and environmentalist persona vectors. Figures (a) and (b) show the comparison between Base Model (without system prompt), Prompt (CRE+ENV), and BILLY (CRE+ENV). Figure (c) and (d) demonstrate the projection of applying (i) only BILLY (CRE), (ii) only BILLY (ENV), and (iii) BILLY (CRE+ENV) at Layer 20.

indicates that MRP fails to consistently induce all intended personas in the model's activations.

In stark contrast, the projections of BILLY (CRE + ENV) consistently being positive on both the v_{CRE} (Figure 4a) and v_{ENV} (Figure 4b). By directly manipulating the latent space, BILLY successfully co-activates multiple personas, validating its superior controllability and interpretability for complex, multi-faceted persona generation.

Efficacy of Persona Vectors Merging. Our analysis confirms the efficacy and selectivity of the individual persona vectors. As shown in Figure 4(c), steering with only BILLY (CRE) induces a strong, sustained projection onto the v_{CRE}, while having a negligible effect on the v_{ENV}. Conversely, BILLY (ENV) in Figure 4(d) strongly activates the environmentalist axis with minimal projection onto the creative one. This demonstrates that the extracted vectors are highly effective, precisely controlling their intended semantic concepts.

BILLY (CRE + ENV) reveals a more complex and interesting interaction. Figure 4c and Figure 4d show that BILLY (CRE + ENV) successfully co-activates both personas, maintaining a positive projection on both v_{CRE} and v_{ENV}. However, this co-activation does not translate linearly to the output response. For instance, while the projection of BILLY (CRE + ENV) on v_{CRE} is lower than that of BILLY (CRE) alone in Figure 4c, its Originality score is actually higher in Figure 2.

This suggests an interesting effect just as we discuss in Section 5.3, and Section G: the environmentalist vector provides substantive content that the creative professional vector can then reframe, leading to a richer and more original output. Besides the projection analysis, we also analyze the trait expression of output response in Section H.

6 Conclusion

We introduce BILLY, an efficient activation steering method that enhances LLM creativity while avoiding the high inference time and costs of multi-LLM systems. Extensive experiments on four benchmarks show that our method BILLY matches or surpasses strong creative baselines with significantly reduced inference time and costs. Furthermore, our qualitative analysis shows that different persona vectors influence distinct aspects of generation, such as style versus content. Via activation projections, we suggest that BILLY offers superior controllability compared to prompting methods over both the model's internal state and its output.

Limitations

The main contributions of our work include: (1) demonstrating that the creative benefits of multi-agent collaboration can be effectively captured within a single model via activation steering, (2) significantly reducing the inference time and token cost for creativity tasks , and (3) revealing the multiple role-play prompting is less controllable and interpretable than activation steering with a composite persona vector by several analyses.

While we recognize that developing a more sophisticated framework for vector composition that moves beyond simple averaging to predictably manage these non-linear interactions is a key direction for advancing controllable AI, such a method is not directly proposed in this work. Future research can build on our findings by developing more advanced composition techniques. These could include learning task-specific weights for each persona vector or designing mechanisms that explicitly model the functional roles to achieve more precise control. Such advancements would support the development of more generalizable and robustly control-